

Cambridge (CIE) IGCSE Additional Maths



Your notes

Permutations & Combinations

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Your notes

Permutations

Arrangements

How many ways can n different objects be arranged?

- When arranging **different** objects in a row, consider how many of the objects can go in the first position, how many can go in the second and so on
 - For example, if $n = 2$ there are two options for the first position and then there will only be one object left to go in the second position
 - So a total of $2 \times 1 = 2$ possible arrangements
 - To arrange the letters A and B we have
 - AB and BA
 - For example, if $n = 3$ there are three options for the first position and then there will be two objects for the second position and one left to go in the third position
 - So a total of $3 \times 2 \times 1 = 6$ possible arrangements
 - To arrange the letters A, B and C we have
 - ABC, ACB, BAC, BCA, CAB and CBA
- For n objects there are n options for the first position, $n - 1$ options for the second position and so on until there is only one object left to go in final position
 - The number of ways of arranging different objects is $n \times (n - 1) \times (n - 2) \times \dots \times 2 \times 1$



Worked Example

By considering the number of options there are for each letter to go into each position, find how many distinct arrangements there are of the letters in the word MATHS.

There are 5 different letters in the word MATHS, so there are 5 letters for the first space, then there will be four for the second, three for the third and so on.

$$5 \times 4 \times 3 \times 2 \times 1$$

120 distinct arrangements



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Factorials

What are factorials?

- Factorials are a type of mathematical operation (just like +, -, ×, ÷)
- The symbol for factorial is !
 - So to take a factorial of any non-negative integer, n , it will be written $n!$. And pronounced ' n factorial'
- The factorial function for any integer, n , is $n! = n \times (n-1) \times (n-2) \times \dots \times 2 \times 1$
 - For example, 5 factorial is $5! = 5 \times 4 \times 3 \times 2 \times 1$
- The factorial of a negative number is not defined
 - You cannot arrange a negative number of items
- $0! = 1$
 - There are no positive integers less than zero, so zero items can only be arranged once
- Most normal calculators cannot handle numbers greater than about 70!, experiment with yours to see the greatest value of X such that your calculator can handle $X!$

How are factorials and arrangements linked?

- The number of arrangements of n different objects is $n!$
 - Where $n! = n \times (n-1) \times (n-2) \times \dots \times 2 \times 1$

What are the key properties of using factorials?

- Some important relationships to be aware of are:
 - $n! = n \times (n-1)!$
 - Therefore

$$\frac{n!}{(n-1)!} = n$$

- $n! = n \times (n-1) \times (n-2)!$
 - Therefore



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$$\frac{n!}{(n-2)!} = n \times (n-1)$$

- Expressions with factorials in can be simplified by considering which values cancel out in the fraction
 - Dividing a large factorial by a smaller one allows many values to cancel out

$$\frac{7!}{4!} = \frac{7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{4 \times 3 \times 2 \times 1} = 7 \times 6 \times 5$$



Examiner Tips and Tricks

- Arrangements and factorials are tightly interlinked with permutations and combinations
- Make sure you fully understand the concepts in this revision note as they will be fundamental to answering perms and combs exam questions!



Worked Example

(i) Show, by writing $8!$ and $5!$ in their full form and cancelling, that

$$\frac{8!}{5!} = 8 \times 7 \times 6$$

(ii) Hence, simplify $\frac{n!}{(n-3)!}$

(i) Write $5!$ and $8!$ in their full form.

$$5! = 5 \times 4 \times 3 \times 2 \times 1$$

$$8! = 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$$

Write them as a fraction and cancel out common terms

$$\frac{8!}{5!} = \frac{8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{5 \times 4 \times 3 \times 2 \times 1}$$



$$\frac{8!}{5!} = 8 \times 7 \times 6$$

(ii) 'Hence' means using the information in the previous question. Let n be 8

$$n = 8, \quad \frac{n!}{(n-3)!} = \frac{8!}{(8-3)!} = \frac{8!}{5!} \quad \text{and} \quad \frac{8!}{5!} = 8 \times 7 \times 6$$

Then $5 = n - 3$, $6 = n - 2$, $7 = n - 1$

$$\frac{n!}{(n-3)!} = n(n-1)(n-2)$$

Permutations

Are permutations and arrangements the same thing?

- Mathematically speaking yes, a **permutation** is the number of possible arrangements of a set of objects when the **order** of the arrangements matters
- A permutation can either be finding the number of ways to arrange n items or finding the number of ways to arrange r out of n items
- The number of permutations of n **different** items is $n! = n \times (n-1) \times (n-2) \times \dots \times 2 \times 1$
 - For 5 **different** items there are $5! = 5 \times 4 \times 3 \times 2 = 120$ permutations
 - For 6 **different** items there are $6! = 6 \times 5 \times 4 \times 3 \times 2 = 720$ permutations
 - It is easy to see how quickly the number of possible permutations of different items can increase
 - For 10 different items there are $10! = 3\,628\,800$ possible permutations

How do we find r permutations of n items?

- If we only want to find the number of ways to arrange a few out of n different objects, we should consider how many of the objects can go in the first position, how many can go in the second and so on
- If we wanted to arrange 3 out of 5 different objects, then we would have 3 positions to place the objects in, but we would have 5 options for the first position, 4 for the second and 3 for the third
 - This would be $5 \times 4 \times 3$ ways of permuting 3 out of 5 different objects
 - This is equivalent to $\frac{5!}{2!} = \frac{5!}{(5-3)!}$



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- If we wanted to arrange 4 out of 10 different objects, then we would have 4 positions to place the objects in, but we would have 10 options for the first position, 9 for the second, 8 for the third and 7 for the fourth
 - This would be $10 \times 9 \times 8 \times 7$ ways of permutating 4 out of 10 different objects
 - This is equivalent to $\frac{10!}{6!} = \frac{10!}{(10-4)!}$
- If we wanted to arrange r out of n different objects, then we would have r positions to place the objects in, but we would have n options for the first position, $(n-1)$ for the second, $(n-2)$ for the third and so on until we reach $(n-(r-1))$
 - This would be $n \times (n-1) \times \dots \times (n-r+1)$ ways of permutating r out of n different objects
 - This is equivalent to $\frac{n!}{(n-r)!}$
- The function $\frac{n!}{(n-r)!}$ can be written as ${}^n\text{P}_r$
 - Make sure you can find and use this button on your calculator
- The same function works if we have n spaces into which we want to arrange r objects, consider
 - for example arranging five people into a row of ten empty chairs

Permutations when two or more items must be together

- If two or more items must stay together within an arrangement, it is easiest to think of these items as 'stuck' together
- These items will become one within the arrangement
- Arrange this 'one' item with the others as normal
- Arrange the items within this 'one' item separately
- Multiply these two arrangements together

Permutations when two or more items cannot be all together

- If **two** items must be **separated** ...
 - consider the number of ways these two items would be together
 - subtract this from the total number of arrangements without restrictions
- If **more than two** items must be separated...



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- consider whether all of them must be **completely separate** (none can be next to each other) or whether they **cannot all be together** (but two could still be next to each other)
- If they **cannot all be together** then we can treat it the same way as separating two items and subtract the number of ways they would all be together from the total number of permutations of the items, the final answer will include all permutations where two items are still together

Permutations when more than two items must be separated

- If the items must all be **completely separate** then
 - lay out the rest of the items in a line with a space in between each of them where one of the items which cannot be together could go
 - remember that this could also include the space before the first and after the last item
 - You would then be able to fit the items which cannot be together into any of these spaces, using the r permutations of n items rule $\left({}^n P_r \right)$
- You do not need to fill every space

Permutations when two or more items must be in specific places

- Most commonly this would be arranging a word where specific letters would go in the first and last place
- Or arranging objects where specific items have to be at the ends/in the middle
 - Imagine these specific items are stuck in place, then you can find the number of ways to arrange the rest of the items around these 'stuck' items
- Sometimes the items must be grouped
 - For example all vowels must be before the consonants
 - Or all the red objects must be on one side and the blue objects must be on the other
 - Find the number of permutations within each group separately and multiply them together
 - Be careful to check whether the groups could be in either place
 - e.g. the vowels on one side and consonants on the other
 - or if they must be in specific places (the vowels **before** the consonants)
 - If the groups could be in either place then your answer would be multiplied by two
 - If there were n groups that could be in any order then your answer would be multiplied by $n!$



Your notes



Examiner Tips and Tricks

- The wording is very important in permutations questions, just one word can change how you answer the question
- Look out for specific details such as whether three items must all be separated or just cannot be all together (there is a difference)
- Pay attention to whether items must be in alternating order (e.g. red and blue items must alternate, either RBRB... or BRBR...) or whether a particular item must come first (red **then** blue and so on)
- If items should be at the ends, look out for whether they can be at either end or whether one must be at the beginning and the other at the end



Worked Example

a) How many ways are there to rearrange the letters in the word EXAMS if the E and the S must be at each end?

'Stick' the E and the S at each end. Be careful here, the question says 'at each end' so the E and the S could each go at either end.

E _ _ _ S

or

S _ _ _ E

Find the number of arrangements of the remaining letters.

$$3! = 3 \times 2 \times 1$$

Multiply by 2 as there were two positions the E and the S could go in.

$$2 \times 3! = 2 \times 3 \times 2 \times 1$$



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(b) Find the number of distinct four digit codes that can be made from the digits 1, 2, 3, 4, 5, 6, 7, 8, 9, 0, where none of the digits can be used more than once.

There are ten numbers available and we want to find the number of arrangements of four of them.

There are 10 numbers for the first position, 9 for the second, 8 for the third and 7 for the fourth.

$$10 \times 9 \times 8 \times 7$$

This is the same as ${}^{10}P_4$, ($n = 10$, $r = 4$).

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Your notes

Combinations

Combinations

What is the difference between permutations and combinations?

- A **combination** is the number of possible arrangements of a set of objects when the **order** of the arrangements **does not matter**
 - On the other hand a permutation is when the order of arrangement **does matter**
- A combination will be finding the number of ways to **choose** r out of n items
 - The order in which the r items are chosen is not important
 - For example if we are choosing two letters from the word CAB, AB and BA would be considered the same combination but different permutations

How do we find r combinations of n items?

- If we want to find the number of ways to **choose** 2 out of 3 different objects, but we don't mind the order in which they are chosen, then we could find the number of **permutations of 2 items from 3** and then divide by the number of ways of arranging each combination
 - For example if we want to choose 2 letters from A, B and C
 - There are 6 permutations of 2 letters:
 - AB, BA, AC, CA, BC, CB
 - For each combination of 2 letters there are 2 (2×1) ways of arranging them
 - (for example, AB and BA)
 - So divide the total number of permutations (6) by the number of ways of arranging each combination (2) to get 3 combinations
- If we want to find the number of ways to **choose** 3 out of 5 different objects, but we don't mind the order in which they are chosen, then we could find the number of **permutations of 3 items from 5** and then divide by the number of ways of arranging each combination
 - For example if we want to choose 3 letters from A, B, C, D and E
 - There are 60 permutations of 3 letters:
 - ABC, ACB, BAC, BCA, CAB, CBA, ABD, ADB, etc



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- For each combination of 3 letters there are 6 ($3 \times 2 \times 1$) ways of arranging them (for example, ABC, ACB, BAC, BCA, CAB and CBA)
- So divide the total number of permutations (60) by the number of ways of arranging each combination ($3! = 6$) to get 10 combinations
- If we want to find the number of ways to **choose** r items out of n different objects, but we don't mind the order in which they are chosen, then we could find the number of **permutations of r items from n** and then divide by the number of ways of arranging each combination
- Recall that the formula for r permutations of n items is

$${}^n P_r = \frac{n!}{(n-r)!}$$

- This would include $r!$ ways of repeating each combination
- The formula for r **combinations** of n items is

$$\frac{{}^n P_r}{r!} = \frac{n!}{(n-r)! r!}$$

- The function $\frac{n!}{(n-r)! r!}$ can be written as ${}^n C_r$ or $\binom{n}{r}$ and is often read as ' n choose r '
 - Make sure you can find and use this button on your calculator
- The formulae for permutations and combinations satisfy the following relationship:

$${}^n C_r = \frac{{}^n P_r}{r!}$$

What do I need to know about combinations?

- The formula ${}^n C_r = \frac{n!}{(n-r)! r!}$ is also known as a **binomial** coefficient
- ${}^n C_n = {}^n C_0 = 1$
 - It is easy to see that there is only one way of arranging n objects out of n and also there can only be one way of arranging 0 objects out of n
 - By considering the formula for this, it reinforces the fact that $0!$ Must equal 1



Your notes

- The binomial coefficients are symmetrical, so ${}^nC_r = {}^nC_{n-r}$

- This can be seen by considering the formula for nC_r

$${}^nC_{n-r} = \frac{n!}{(n-r)!(n-(n-r))!} = \frac{n!}{r!(n-r)!} = {}^nC_r$$

How do I know when to multiply or add?

- Many questions will ask you to find combinations of a group of different items from a bigger group of a specified number of those different items
 - For example, find the number of ways five questions could be chosen from a bank of twenty different pure and ten different statistics questions
 - The hint in this example is the word 'chosen', this tells you that the order in which the questions are chosen doesn't matter
- Sometimes questions will have restrictions,
 - For example there should be three pure and two statistics chosen from the bank of questions,
 - Or there must be at least two pure questions within the group
- If unsure about whether to add or multiply your options, ask yourself if A **and** B are both needed, or if A **or** B is needed
 - Always **multiply** if the answer is **and**, and **add** if the answer is **or**
 - For example if we needed exactly three pure **and** two statistics questions we would find the amount of each and multiply them
 - If we could have either five statistics **or** five pure questions we would find them separately and add the answers
- Probabilities can be found with combinations questions by finding the number of options a selection can be made in a particular way and dividing that by the total number of options



Examiner Tips and Tricks

- It is really important that you can tell whether a question is about permutations or combinations
 - Look out for key words such as **arrange** (for permutations) or **choose** or **select** (for combinations)

- Don't be confused if a question asks for the number of **ways**, this could be for either a permutations or a combinations question
 - Look out for other clues



Worked Example

Oscar has to choose four books from a reading list to take home over the summer. There are four fantasy books, five historical fiction books and two classics available for him to choose from. In how many ways can Oscar choose four books if he decides to have:

- (i) two fantasy books and two historical fictions?
- (ii) at least one of each type of book?
- (iii) at least two fantasy books?



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Your notes

- (i) Choosing two fantasy from four: 4C_2
 Choosing two historical fiction from five: 5C_2
 Total options = ${}^4C_2 \times {}^5C_2 = 6 \times 10$

We need fantasy **and** historical fiction so **multiply** the options

60 options

- (ii) Oscar must have two of one and one of each of the others
 Let F represent a fantasy book, H represent historical fiction
 and C represent a classic

	Has	4	5	2	
	Can choose	F	H	C	
These are the choices he can make		1	1	2	And ${}^4C_1 \times {}^5C_1 \times {}^2C_2 = 4 \times 5 \times 1 = 20$ ${}^4C_1 \times {}^5C_2 \times {}^2C_1 = 4 \times 10 \times 2 = 80$ ${}^4C_2 \times {}^5C_1 \times {}^2C_1 = 6 \times 5 \times 2 = 60$
		1	2	1	
		2	1	1	

These are the options he can choose from

He can choose from one of these options so we add them:

$$20 + 80 + 60 = 160$$

'Or' so add

160 options

- (iii) He could have two, three or four fantasy books:

	Has	4	5	2
	Can choose	F	H	C
Use a system to make sure you don't forget any Check that all the options add up to 4		2	2	0
		2	1	1
		2	0	2
		3	1	0
		3	0	1
		4	0	0

Always check that these numbers are not bigger than the possible options

${}^4C_2 \times {}^5C_2 \times {}^2C_0 = 60$ ${}^4C_2 \times {}^5C_1 \times {}^2C_1 = 60$ ${}^4C_2 \times {}^5C_0 \times {}^2C_2 = 6$ ${}^4C_3 \times {}^5C_1 \times {}^2C_0 = 20$ ${}^4C_3 \times {}^5C_0 \times {}^2C_1 = 8$ ${}^4C_4 \times {}^5C_0 \times {}^2C_0 = 1$	}	Any one of these could be chosen this is 'OR' so add them
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$$\text{Total number of options} = 60 + 60 + 6 + 20 + 8 + 1 = 155$$

155 options

Alternative method: it doesn't matter whether the other books are historical fiction or classics so we could just consider these as 'not fantasy', (N)

4 7
F N

$$\begin{array}{lll} 2 & 2 & {}^4C_2 \times {}^7C_2 = 6 \times 21 = 126 \\ 3 & 1 & {}^4C_3 \times {}^7C_1 = 4 \times 7 = 28 \\ 4 & 0 & {}^4C_4 \times {}^7C_0 = 1 \times 1 = 1 \end{array}$$

$$126 + 28 + 1 = 155 \text{ options}$$

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Your notes



Your notes

Problem Solving with Permutations & Combinations

Problem Solving with Permutations & Combinations

How will I know whether to use Permutations or Combinations?

- In **permutation** problems, the **order** of arrangement matters
 - For example, guessing who's going to win first, second and third prize in a competition
- In **combination** problems, the **order** of arrangement doesn't matter
 - For example, guessing the top three high scorers from a group of students
- Look out for **keywords** in a problem
 - **Permutation** questions will most likely use the words '**arrange**', '**order**', '**sequence**'
 - **Combination** questions will most likely use the words '**select**', '**choose**', '**group**'
- Pay close attention to the **context** of the problem to determine if the order matters or not

How should I approach perms and combs questions?

- Identify whether the question is a permutations or a combinations question
- Identify the total number of options available (n) and the number to be arranged or selected (r)
- Consider whether objects taken from each list can be repeated or not
 - For example a four digit PIN from ten numbers where each number can be used repeatedly would be $10 \times 10 \times 10 \times 10$
 - There are 10 options for the first and ten options for the second number and so on
 - If the numbers could only be used once then the number of options for each digit would **reduce** with each digit
 - There are 10 options for the first, nine options for the second, eight for the third and so on
- Look out for any limitations or conditions in the problem
 - Do some items need to be kept together or separate?
 - Does a code or a number need to begin or end in a particular value?

How will I know whether to add or multiply?



Your notes

- If a question requires you to choose an item from one list **AND** an item from another list you should **multiply** the number of options in each list
- In general if you see the word 'AND' you will most likely need to 'MULTIPLY'
 - For example if you are choosing a pen and a pencil from 4 pens and 5 pencils:
 - You can choose 1 item from 4 pens AND 1 item from 5 pencils
 - You will have 4×5 different options to choose from
- If a question requires you to choose an item from one list **OR** an item from another list you should **add** the number of options in each list
- In general if you see the word 'OR' you will most likely need to 'ADD'
 - For example if you are choosing a pen or a pencil from 4 pens and 5 pencils:
 - You can choose 1 item from 4 pens OR 1 item from 5 pencils
 - You will have $4 + 5$ different options to choose from
- If a question has a restriction, such as 'four items containing at least one yellow and no more than three reds' you will need to
 - Find all of the possibilities (one yellow **and** three reds **or** two yellows **and** two reds etc)
 - Multiply for each of the 'and' statements (one yellow **and** three reds)
 - Add together each of the 'or' statements at the end (one yellow and three reds **or** two yellows and two reds etc)



Examiner Tips and Tricks

- Read the question twice, taking note of the key areas outlined in this revision note and looking out for any keywords
- If you're unsure, try writing down a few of the possible options to look for clues



Worked Example

A password must be 5 characters long and consist of 2 letters from the letters A, B, C, D, and E followed by 3 digits from the numbers 0, 1, 2, 3, 4, 5, 6, 7, 8, 9. Each letter or number may only be used once within the password. Find how many ways the password can be formed.



Your notes

There are no keywords to help us here so we must consider to context of the question. The context is about a password so the **order** of the arrangement matters, this is a **permutations** question.

The 2 letters must be **followed** by the three digits, so we must treat these separately.

For the first letter there are five possible options (A, B, C, D or E) For the second letter there are four possible options as no letter can be repeated.

$${}^5P_2 = (5 \times 4)$$

For the first number there are ten possible options (0, 1, 2, 3, 4, 5, 6, 7, 8, 9) For the second number there are nine possible options as no number can be repeated. For the third number there are eight possible options.

$${}^{10}P_3 = (10 \times 9 \times 8)$$

We want two letters **AND** three numbers so we will multiply the options.

$${}^5P_2 \times {}^{10}P_3$$

14400 passwords